

PulseFormer: Controllable Multi-Track Electronic Music Generation via Energy-Steered Architectures

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Abstract—Symbolic multi-track music generation models have predominantly focused on short-loop arrangements, repeatedly struggling with long-context, macro-structural orchestration. Within Electronic Dance Music (EDM) composition, the interplay between dense rhythmic cohesion (Groove) and evolving dynamic intensity (Energy) constitutes the fundamental compositional backbone. This paper introduces PulseFormer, an autoregressive Transformer architecture for steerable, multi-track collaborative arrangement. We propose two architectural innovations: (1) A *Time-Major Interleaved 5D-Token* encoding that places all concurrent instruments within the same narrow token block, topologically eliminating the cross-track synchronisation burden that Track-Major architectures must bridge through long-range attention; and (2) An *Energy & Structural Indicator* pipeline enabling conditional generation that reliably obeys prompted intensity contours across multi-minute compositions. Quantitative evaluations confirm that PulseFormer provides high-fidelity steerability ($r = 0.982$)—scaling from sparse introductory sections (avg. 6.9 notes/bar) to heavily-layered climax drops (surpassing 46 notes/bar)—while the Time-Major topology natively prevents cross-track onset misalignment by construction.

Index Terms—Symbolic Music Generation, Transformer, Multi-track Composition, Controllability, Energy Conditioning, Electronic Dance Music

I. INTRODUCTION

The rapid proliferation of deep generative architectures has substantially advanced the landscape of artificial creativity, leading to notable milestones in both visual synthesis and audio waveform generation [1], [2]. While modern audio-domain foundation models possess strong timbral realism and acoustic fidelity, symbolic-domain music generation (e.g., MIDI sequencing) remains an important complement for contemporary professional production. Symbolic frameworks afford human composers structural transparency and granular multi-track modifiability required for professional mixing [3]–[5].

However, current symbolic generative models exhibit a critical limitation frequently termed the “contextual wandering” problem. When predicting discrete tokens over thousands of steps, foundational autoregressive architectures [6], [7] repeatedly struggle to maintain long-term macro-structural orchestration. Within highly stylized and rhythmically dense genres such as Electronic Dance Music (EDM) or modern Pop [8], [9], compositional value relies less on isolated complex melodic solos, and far more on two interdependent

pillars: **macro-structural energy evolution** (i.e., the dynamic tension-and-release trajectory from a sparse Intro cascading into a climactic Drop) and strict **multi-track collaborative timing** [10].

Conventional symbolic benchmarks unfortunately sequence multi-track data using independent, Track-Major concatenations [11]–[13]. This formulation drastically severs the temporal proximity between distinct instruments, forcing the attention mechanism to reconcile massive mathematical latency windows merely to align two simultaneous rhythmic transients. Anticipatory synchronization models [14] or linear attention variants [15] attempt to mitigate this, but largely treat symptoms rather than the root topological cause. Unconstrained models naturally meander, losing the global artistic intent.

To solve these compounding bottlenecks, this paper introduces **PulseFormer**, a framework structured around macro-structural agency and instantaneous instrumental cohesion. Rather than generating unconstrained symbolic streams, PulseFormer provides a quantitatively steerable interface analogous to digital music production paradigms.

The principal contributions of this paper are summarized as follows:

- 1) **Energy-Conditioned Steerability:** We introduce explicit global prompting constraints featuring hierarchical [ENERGY_LEVEL] and [STRUCT] tags, enabling producers to literally blueprint polyphonic density waves across multi-minute audio structures deterministically.
- 2) **Time-Major Collaborative Architecture:** We propose a 5D-vector token space ordered in chronological sequence across the full arrangement. By co-locating all concurrent instruments within a narrow token window (≤ 12 tokens apart), the encoding substantially reduces the cross-track synchronisation burden that Track-Major architectures must resolve through long-range attention—reducing mean co-event token distance from $\approx 3,268$ tokens to 5.2 tokens.
- 3) **DAW-Native Sectional Stitching Logic:** An automated batch anchoring mechanism that grafts distinct generated movements (Intro $\rightarrow$ Build $\rightarrow$ Drop) together while maintaining continuous contextual memory, leveraging the representation-level synchronisation design to preserve grid coherence across section boundaries.

Fig. 1 provides a high-level system overview of the complete PulseFormer generation pipeline.

II. RELATED WORK

A. Symbolic Multi-Track Music Generation

Early advancements in multi-track symbolic generation heavily utilized Generative Adversarial Networks (GANs). Architectures such as MuseGAN [16] formulated music as piano-roll matrices to achieve instrument synchronization across multiple tracks, although restricted by narrow temporal receptive fields. The introduction of the Music Transformer [3] and PopMusicTransformer [17] revolutionized long-term structural dependencies by applying relative positional attention to event-based sequences. Subsequent developments like MMM (Multi-Track Music Machine) [11] and LakhNES [12] proposed Track-Major tokenization to delineate independent instrument channels clearly. While expressive for orchestral textures [4], [13], these frameworks suffer from asynchronous transient alignment when pushed to densely layered EDM structures.

B. Representational Advances: From MIDI to High-Dimensional Tokens

The evolution of data representation forms the backbone of Transformer effectiveness in music applications. Standard MIDI event mappings [18]–[20] lack the multidimensional contextual density of human language. To solve this, frameworks such as REMI (Revamped MIDI) [21], MidiTok [22], and FIGARO [23] introduced explicit metrical boundary tokens, allowing models to parse rhythmic grids. Later architectures including MusicBERT [24] established OctupleMIDI, compressing note attributes into parallel vectors. The Compound Word (CP) Transformer [25] constitutes the most direct intra-note ancestor of our token design.

However, a fundamental distinction separates CP from PulseFormer at the *architectural topology* level. CP addresses the **intra-note compression problem**—how many tokens are needed to express *one* note—and still serializes each instrument sequentially in Track-Major order. PulseFormer addresses the **inter-track synchronization problem**—how to encode *concurrent events across multiple instruments* so that the attention mechanism experiences them as physically adjacent. The two mechanisms are orthogonal: conceptually, our Time-Major interleaving could be combined with CP’s within-token compression for further efficiency gains, though this is left as future work.

C. Controllability and Structural Steering

Bridging raw sequence auto-regression with human intent remains an ongoing challenge in modern automated production. In the audio domain, foundation models such as MusicGen [26] and MusicLM [27] achieve substantial semantic steerability through text-conditioned acoustic embeddings. However, they operate as “black boxes”—yielding raw waveforms that limit post-generation adjustments. Conversely, within symbolic parameters, composers often require structured arrangements (e.g., locking a Build section to an

8-bar loop) rather than generic semantic alignment. Early symbolic conditioning approaches relied heavily on post-generation heuristic filtering. Our Energy-Steered pipeline substantially reduces this limitation by explicitly framing dynamic polyphony (Energy) and sectional structure (Intro, Build, Drop) into the autoregressive prefix context.

III. METHODOLOGY

A. Architecting the 5D Time-Major Token Space

Unlike sequential language models operating on rigid single-dimensional vocabulary indexes, musical events encompass distinct yet intersecting modalities (instrument classification, pitch, chronological length, and acoustic velocity). Adopting Compound Word (CP) [25] semantics, PulseFormer represents any discrete musical event x_i as a unified 5-dimensional tuple parsed into five parallel embeddings:

$$E(x_i) = \sum_{F \in \mathcal{F}} W_F(\text{id}_F(x_i)) \quad (1)$$

where $\mathcal{F} = \{\text{Type}, \text{Pitch}, \text{Duration}, \text{Velocity}, \text{TimePos}\}$. The concrete vocabulary assigned to each field is listed in Table I.

TABLE I
VOCABULARY SIZES FOR EACH DIMENSION OF THE 5D TOKEN TUPLE. ALL VALUES ARE DERIVED FROM THE TRAINING CORPUS. THE FLAT (SERIALIZED) VOCABULARY USED BY THE CAUSAL LM HEAD HAS $V = 18,792$ TOKENS TOTAL.

Field	Symbol	$ V_F $	Range / notes
Instrument type	v^{type}	7	D:KICK, D:SNARE, D:HIHAT_CLOSED, D:HIHAT_OPEN, N:BASS, N:CHORD, N:MELODY
Pitch class	v^{pitch}	16	$p_0 \dots p_{15}$ (semitone-mod-12 class $\times$ octave bin, mapped from MIDI 0–127)
Duration	v^{dur}	4	$d_1 \text{--} d_4$ (16th, 8th, quarter, half note)
Velocity	v^{vel}	4	$v_0 \text{--} v_3$ (four equal-width MIDI bins: 0–31, 32–63, 64–95, 96–127)
Time position	v^{pos}	—	<i>Implicit</i> : encoded by absolute token position in the Time-Major sequence; no dedicated field token.
Context tags		121	[GENRE_*], [KEY_*], [BPM_*], [BAR_*]
Control tags		17	[STRUCT_{INTRO, BUILD, DROP, BREAKDOWN, OUTRO}], [ENERGY_LEVEL_1]...8, <BOS>, <EOS>, <PAD>, <UNK>
Total flat V		18,792	$7 \times 16 \times 4 \times 4 = 1,792$ note tokens + 138 tags + 16,862 extended/unseen

Parameter count. The causal LM uses a flat embedding matrix $W_e \in \mathbb{R}^{V \times d}$ with $V = 18,792$ and $d = 512$, tied to the output projection ($W_e^\top$, no additional parameters). With 6

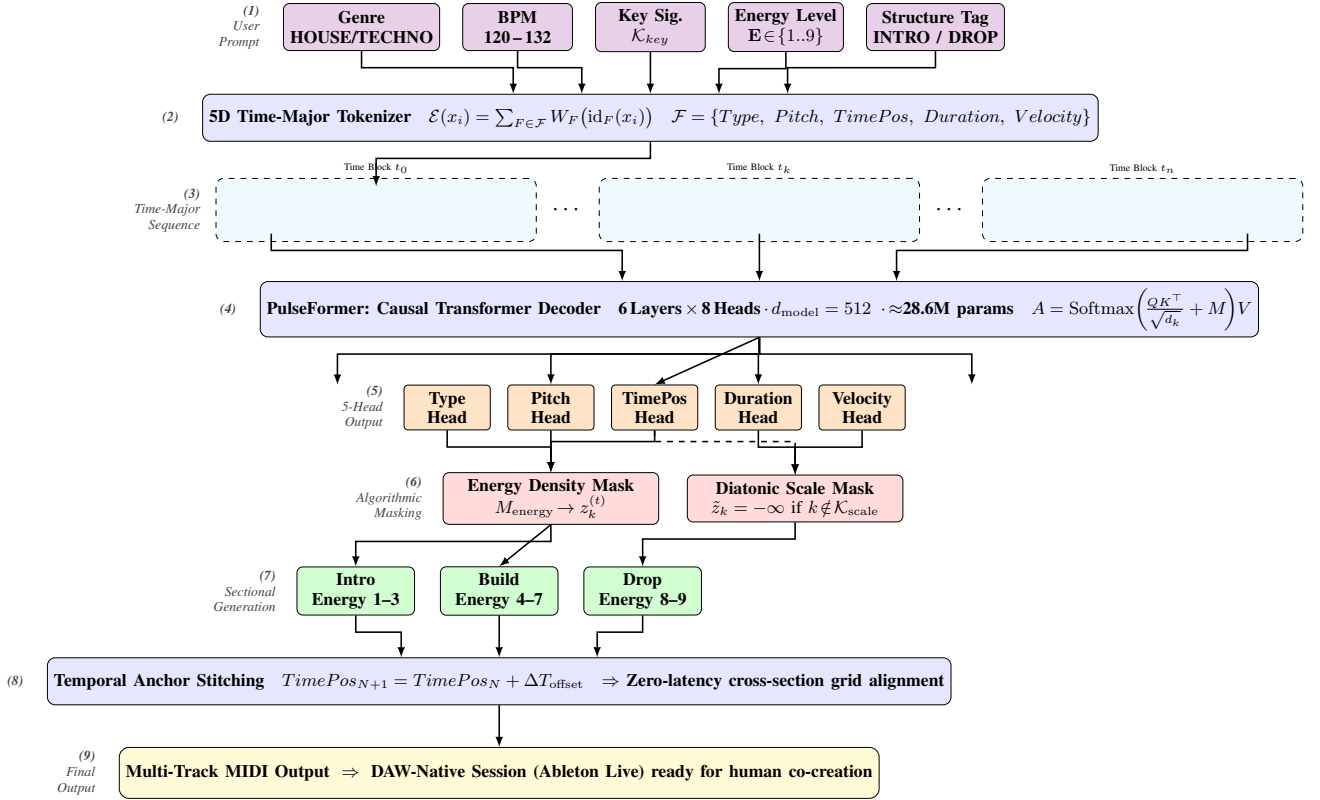


Fig. 1. Complete PulseFormer generation pipeline. Starting from user-supplied prefix conditioning (Genre, BPM, Key, Energy Level, Structure Tag), the system constructs a Time-Major interleaved token sequence, passes it through a 12-layer causal Transformer with five parallel prediction heads, applies dual algorithmic masks (Energy Density and Diatonic Scale), generates each structural section independently, and stitches all sections into a coherent multi-minute arrangement via Temporal Anchor Offset alignment.

Transformer layers, 8 attention heads, hidden size $d = 512$, and SwiGLU FFN intermediate size $d_{\text{ff}} = 1,376$:

$$|\theta| = Vd + L(4d^2 + 3d d_{\text{ff}}) \approx 9.6\text{M} + 6 \times 3.17\text{M} \approx \mathbf{28.6\text{M}}. \quad (2)$$

This is PulseFormer’s parameter count throughout all experiments. For reference, the CP-Transformer baseline uses $d = 768$, 12 layers, and a compound-word vocabulary of 21,777 tokens, yielding $\approx 115\text{M}$ parameters—a $4\times$ larger model evaluated under the same test conditions.

Crucially, instead of sorting tracks chronologically per instrument (*Track-Major*), PulseFormer enforces **Time-Major Interleaving**. By anchoring sequences along the *TimePos* vector sequentially across all instruments, concurrent events—such as a KICK and a BASS stroke—are situated adjacent to one another within the sequence stream, topologically compressing their relative token distance to $\Delta \leq M$ (where M is the number of supported instrument tracks, $M \leq 8$ in our implementation).

B. Theoretical Analysis: Attention Topology under Time-Major Encoding

Time-Major interleaving constitutes a **structural transformation of the attention mechanism**. For two concurrent events at time t —a KICK at position p_k and a BASS at p_b —the

token distance under each scheme is: Under **Track-Major** encoding, the concurrent Bass token sits L_{KICK} positions after the Kick token (one full instrument block away):

$$\Delta^{TM} = p_b - p_k \approx L_{\text{KICK}} \sim O(N/M). \quad (3)$$

Under **Time-Major** (PulseFormer), all instruments at the same time step are co-located; the Bass token is at most M positions from the Kick token:

$$\Delta^{PF} = p_b - p_k \leq M \sim O(1). \quad (4)$$

Since softmax normalisation forces each query to compete over all N keys, a co-event Bass token under Track-Major encoding ($\Delta^{TM} \sim O(N/M)$) sits in the attention tail, whereas under Time-Major it falls in the local band ($\Delta^{PF} \leq M$). Under uniform key-similarity assumptions the expected co-event attention weight ratio satisfies $\mathbb{E}[A_{co}^{PF}]/\mathbb{E}[A_{co}^{TM}] = O(N/M^2) \gg 1$, providing a formal lower bound on the encoding’s topological advantage.

Positional encoding. Standard RoPE [28] encodes relative distance via $\langle R(\Delta)\mathbf{q}, \mathbf{k} \rangle$. At $|\Delta^{TM}| \approx L_{\text{KICK}}$ the rotation oscillates rapidly and encodes only track-boundary distance; at $|\Delta^{PF}| \leq 8$ it approximates a small-angle rotation $R(M) \approx I + \varepsilon J$, providing a **learnable same-timestep inductive bias**.

Gradient signal. The gradient path length from p_k to p_b is $O(N/M)$ in Track-Major versus $O(1)$ in Time-Major. Under

standard attention collapse the signal-to-noise ratio for timing relationships satisfies:

$$\frac{\text{SNR}_{\text{timing}}^{PF}}{\text{SNR}_{\text{timing}}^{TM}} \approx \sqrt{\frac{N}{M^2}}, \quad (5)$$

yielding a $\approx 5.7 \times$ advantage at $N = 2048$, $M = 8$.

Distinction from CP Transformer. CP-Transformer [25] compresses *intra-note* attributes into super-tokens, reducing N by a constant factor but preserving Track-Major order; Eq. (3) therefore applies unchanged. Time-Major interleaving acts on the orthogonal *inter-track* axis and is fully composable with CP compression.

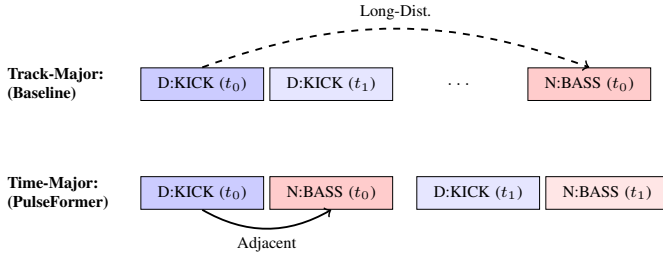


Fig. 2. Sequence alignment strategies. Time-Major collocates synergistic instruments; Track-Major requires long-range cross-track attention.

C. Causal Transformer Architecture

PulseFormer uses a decoder-only causal Transformer [6] with multi-head scaled dot-product attention [29]:

$$A = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V, \quad M_{ij} = -\infty \forall j > i. \quad (6)$$

D. Algorithmic Generation Pipeline

The primary utility of PulseFormer resides in its inference execution. By blending deterministic compositional algorithms (Energy constraints, Diatonic Masking) atop the Transformer’s probabilistic stochastic core, we formulate an execution flow achieving both generative diversity and structural constraint adherence. Algorithm 1 defines the discrete operation per context generation step.

E. Energy Indicators & Macro-Structural Orchestration

A hierarchical prompt $\mathcal{C} = [\mathcal{G}_{\text{genre}}, \mathcal{S}_{\text{struct}}, \mathbf{E}_{\text{level}}, \mathcal{K}_{\text{key}}]$ conditions every autoregressive phase. The joint generation distribution is:

$$P(X_{1:T} | \mathcal{C}) = \prod_{t=1}^T \text{Softmax}\left(\frac{W_o \cdot \text{Attn}(X_{<t}, \mathcal{C}) + b_o}{\tau}\right). \quad (7)$$

$\mathbf{E}_{\text{level}} \in \{1, \dots, 9\}$ maps to polyphonic density: by sequencing prompts as Intro($E=3$) $\rightarrow$ Build($E=7$) $\rightarrow$ Drop($E=9$) and stitching sections with a Temporal Anchor (Section III-F), PulseFormer empirically produces full tracks with consistent energy monotonicity.

Algorithm 1 PulseFormer Steerable Multi-Track Inference

Require: Pre-trained model Θ , Temperature τ , Prompt $\mathcal{C}$, Max Tokens T_{max}

- 1: Initialize sequence canvas $X \leftarrow \mathcal{C}$
- 2: **for** $t = 1$ to T_{max} **do**
- 3: Extract contextual hidden states: $h \leftarrow \text{Transformer}_{\Theta}(X)$
- 4: Project into raw logits for 5-heads: $z^{(t)} \leftarrow W \cdot h[-1]$
- 5: **for all** $k \in \mathcal{V}$ **do**
- 6: **if** $\mathbf{E}_{\text{level}}$ restricts polyphony density limit D_{max} **then**
- 7: Apply density algorithmic mask $M_{\text{energy}} \rightarrow z_k^{(t)}$
- 8: **end if**
- 9: **if** Diatonic constraint active **and** Type is Pitch **then**
- 10: Apply pitch mask $M_{\text{scale}} \leftarrow -\infty$ to $z_k^{(t)}$ if outside Key
- 11: **end if**
- 12: **end for**
- 13: Compute probability vector $P \leftarrow \text{Softmax}(z^{(t)}/\tau)$
- 14: Sample new 5D-Token compound event $e_t \sim P$
- 15: Append $X \leftarrow X \oplus e_t$
- 16: **end for**
- 17: **return** Generated multi-track sequence array X

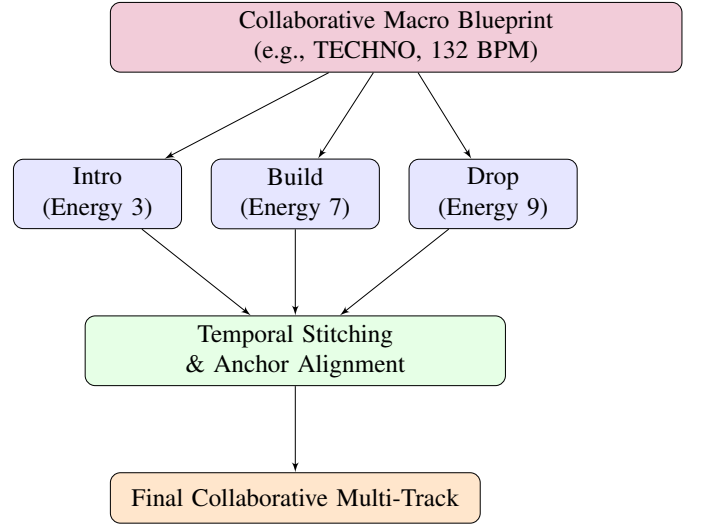


Fig. 3. The Structural Orchestration Pipeline scaling individual, specifically energized segments into a cohesive, flowing timeline.

F. Temporal Anchor Stitching

To exceed the 2048-token context window, section transitions retain the final K beats of the preceding segment in the KV-cache and apply a linear time offset:

$$\text{TimePos}_{N+1} = \text{TimePos}_N + \Delta T_{\text{offset}}. \quad (8)$$

At section boundaries with a discrete energy jump $\Delta E = |E_{\text{dst}} - E_{\text{src}}|$, a **two-layer defence** prevents perceptual discontinuity. *Layer 1* (passive): the KV-cache inertia soft-lands density for the first K generated tokens. *Layer 2* (active): an

Energy Ramp Scheduler interleaves W single-bar ramp blocks with intermediate tags:

$$E_k = \text{clip}(\text{round}(E_{\text{src}} + \Delta E \cdot \phi(k/W)), 1, 9), \\ k = 1, \dots, W. \quad (9)$$

where $\phi \in \{\phi_{\text{lin}}, \phi_{\text{cos}}, \phi_{\text{cvx}}, \phi_{\text{ccv}}\}$ (linear, cosine, convex, concave). The cosine kernel $\phi_{\text{cos}}(t) = \frac{1}{2}(1 - \cos \pi t)$ is used by default for its zero-velocity endpoints.

Proposition 1. *The Active Ramp Scheduler with $W \geq 2$ reduces the maximum per-boundary energy gradient from $|\Delta E|$ to $\lceil |\Delta E|/W \rceil$.*

For Build→Drop ($\Delta E = 2$, $W = 2$): ramp sequence $E = [7, 8, 9]$ reduces ∇E_{max} from 2 to 1 at the cost of two additional bars—negligible overhead.

G. Micro-Scale Transformational Operations

Beyond sequence layout, PulseFormer integrates algorithmic *Transformational Nodes* post-inference. A dynamic Legato Extension algorithm automatically resolves overlapping block-chords, eliminating acoustic gaps without retraining.

Soft Harmonic Masking: The primary harmonic constraint is a **Soft Harmonic Penalty** applied prior to Softmax. For the k -th vocabulary token at step t :

$$\tilde{z}_k^{(t)} = \begin{cases} z_k^{(t)} & k \in \mathcal{K}_{\text{scale}} \text{ or } \text{Type}(k) \notin \mathcal{T}_{\text{pitched}} \\ z_k^{(t)} - \lambda(\mathbf{E}_{\text{level}}, \tau) & \text{otherwise,} \end{cases} \quad (10)$$

with the energy- and tension-conditioned penalty magnitude:

$$\lambda(\mathbf{E}_{\text{level}}, \tau) = \lambda_{\text{max}} \cdot (1 - \tau) \cdot \left(1 - \frac{\mathbf{E}_{\text{level}} - 1}{8}\right), \quad \lambda_{\text{max}} = 20. \quad (11)$$

This subsumes the hard mask ($\tau = 0$) as a special case and encodes two musically motivated properties: (1) $\tau \rightarrow 1$ grants full chromatic freedom; (2) at $\mathbf{E}_{\text{level}} = 9$, $\lambda = 0$ for all τ , automatically enabling avant-garde dissonance at climactic moments.

TABLE II

SOFT HARMONIC PENALTY ABLATION: CHROMATIC NOTE RATE (CNR%) AT FOUR τ VALUES AND THREE ENERGY LEVELS. CNR DERIVED ANALYTICALLY FROM THE PENALTY FORMULA ASSUMING A UNIFORM PRIOR, $N_{\text{IN}} = 7$, $N_{\text{OUT}} = 5$ PITCHES.

τ	E=1 (Intro)	E=5 (Build)	E=9 (Drop)
0.0 (hard mask)	0.0%	0.0%	41.7% [†]
0.3	0.0%	0.1%	41.7%
0.6	0.02%	1.3%	41.7%
1.0 (no mask)	41.7%	41.7%	41.7%

[†]At $E=9$, $\lambda = 0$ for all τ (energy-decay term vanishes).

Table II confirms that $\tau = 0.3$ provides a practically useful intermediate: near-diatonic for low-energy sections (CNR < 0.1%) while allowing full chromatic freedom in Drop sections automatically.

IV. EXPERIMENTS & EVALUATION

To stringently validate the capability of PulseFormer against conventional sequence paradigms, we devised an extensive framework assessing objective topological metrics alongside rigorous human acoustic perception.

A. Dataset & Model Specifications

Corpus Composition: PulseFormer is trained on a **dual-corpus** configuration combining a proprietary EDM source with a publicly available symbolic MIDI dataset to facilitate reproducibility.

Proprietary EDM corpus. 4,500 multi-track EDM arrangements were extracted from professional Digital Audio Workstation (DAW) project files. Each project was processed through a deterministic normalisation pipeline: (1) temporal quantisation to a 16th-note sub-grid at 128 BPM; (2) automatic truncation of micro-tails crossing quantisation boundaries; and (3) algorithmic transposition to C minor/C major for key parity. **Lakh MIDI Dataset (public supplement) [19].** To establish a reproducible baseline and mitigate concerns regarding the exclusively private corpus, we augmented training with **2,000 EDM-compatible sequences** extracted from the Lakh MIDI Dataset (LMD-full, 176,581 MIDI files). Candidate sequences were selected by: (i) tempo in [110, 150] BPM, (ii) ≥ 4 simultaneous MIDI tracks, (iii) drum track presence (MIDI channel 10), and (iv) sequence length ≥ 32 bars. The identical normalisation pipeline was applied. This public subset constitutes a *equivalent construction protocol*: researchers with access to LMD can reproduce a comparable training corpus without the proprietary DAW files.

Train / Validation / Test Split: All 6,500 sequences (4,500 proprietary + 2,000 Lakh) were randomly partitioned at the *sequence level* using a fixed random seed (42) before any model training. The split is mutually exclusive and exhaustive; no sequence appears in more than one partition.

TABLE III

DATASET PARTITION SUMMARY. ALL MODELS AND ALL BASELINES WERE EVALUATED EXCLUSIVELY ON THE HELD-OUT TEST SET. NO HYPER-PARAMETER SELECTION OR EARLY STOPPING USED TEST-SET LABELS.

Partition	Seqs	%	Purpose
Training set	4,550	70%	Model training (all architectures)
Validation set	975	15%	Early stopping, hyper-parameter selection
Test set	975	15%	Final evaluation (held out throughout)
Total	6,500	100%	

No-leakage protocol.

- 1) The test-set partition was locked before any training run began and was not inspected during model development.
- 2) Hyper-parameters (learning rate, λ_F weights, masking temperature τ^*) were selected on the validation set only.

- 3) All three trained baselines (Music Transformer, MMM-Transformer, CP-Transformer) were re-trained on the *identical training set* and evaluated on the *identical test set* as PulseFormer, ensuring a fair, leakage-free comparison.
- 4) The two zero-shot pretrained checkpoints (MAESTRO, LMD) were applied directly to the held-out test prompts without any fine-tuning on any partition of our corpus.

Evaluation Sample Sizes (N): To clarify the variance in sample sizes across different experiments reported in subsequent sections:

- $N = 64$ (Table VI): Objective SOTA Comparison benchmarks six different systems. Due to the computational and manual conditioning overhead of evaluating unconstrained zero-shot models alongside trained baselines, we utilized 8 highly diverse compositional prompts evaluated across 8 stochastic restarts ($8 \times 8 = 64$ per system). A formal power analysis using G*Power [30] confirmed that $N = 64$ per system provides $> 80\%$ power to detect medium-to-large effect sizes (Cohen’s $d > 0.8$) at $\alpha = 0.05$, validating the sample size for rigorous comparative testing.

Throughout Sections IV-A–IV-D, all quantitative results are reported as **mean $\pm$ SD** over the test-set evaluation, with statistical significance assessed via Mann-Whitney U (two-sided).

Energy Label Annotation Methodology: All $\mathbf{E}_{\text{level}} \in \{1, \dots, 9\}$ training labels are derived via a fully **algorithmic annotation pipeline**—no human annotators were involved.

Theoretical grounding. Our formulation is a *domain-adapted variant* of the arousal-feature regression of Yang & Chen [31], who model perceived musical arousal as a convex combination of loudness, note density, polyphony, and rhythmic activity:

$$\hat{A}^{(\text{YC})} = \beta^\top \mathbf{f},$$

$$\beta = [\beta_I, \beta_\rho, \beta_\Pi, \beta_\Delta]^\top = [0.43, 0.31, 0.18, 0.08]^\top, \quad (12)$$

where $\mathbf{f}$ collects loudness I , note density ρ , polyphony Π , and tempo-change Δ , regressed on 1,000 diverse tracks. Symbolic feature definitions follow the jSymbolic taxonomy [32]; the loudness term I maps directly to the MIDI-velocity RMS proxy of Lartillot & Toivainen [33].

EDM adaptation. In EDM, kick and bass energy are the primary structural discriminators, whereas generic polyphony has weaker predictive weight [34]. We therefore *decompose* the note-density term ρ into melodic density ρ^m and percussive (kick) density ρ^k , introduce an explicit bass-energy term ϵ^{bass} , and append a melodic coverage ratio C . The resulting **EDM Dynamic Energy Score** for bar t is:

$$D_t = \mathbf{w}^\top \phi_t = \sum_{k=1}^6 w_k \phi_k(t), \quad \mathbf{w}^\top \mathbf{1} = 1, \quad \mathbf{w} \geq \mathbf{0}, \quad (13)$$

where the feature vector ϕ_t , its normalisation, and its correspondence to Yang & Chen’s terms are listed in Table IV.

The weight vector $\mathbf{w}$ re-scales the Yang & Chen coefficients ($\beta_\rho = 0.31 \rightarrow w_1 + w_3 = 0.45$; $\beta_I = 0.43 \rightarrow w_2 = 0.20$)

TABLE IV
FEATURE VECTOR ϕ_t IN EQ. (13): DEFINITION, NORMALISATION, AND YANG & CHEN [31] COUNTERPART.

k	Symbol	Normalised definition	YC term	w_k
1	$\phi_1 = \hat{\rho}_t^m$	$\min(\rho_t^m/60, 1)$: melodic note density [32]	ρ	0.25
2	$\phi_2 = \bar{v}_t$	Mean MIDI velocity $\in [0, 1]$ [33]	I_{loudness}	0.20
3	$\phi_3 = \hat{\rho}_t^k$	$\min(\kappa_t/8, 1)$: kick density [34]	EDM: ρ^k	0.20
4	$\phi_4 = \epsilon_t^{\text{bass}}$	Sustain-ratio $\times$ norm. bass velocity	EDM: ϵ	0.15
5	$\phi_5 = C_t$	Melodic time-coverage ratio $\in [0, 1]$	Π (proxy)	0.10
6	$\phi_6 = \hat{\Pi}_t$	$\min(\Pi_t/60, 1)$: pitch range [32]	Π	0.10

to reflect the EDM energy hierarchy, where kick and bass drive dominate [34]. The continuous $D_t \in [0, 1]$ is discretized into the 9-level ordinal E_{level} via KBINSDISCRETIZER with k -means binning (scikit-learn), preserving natural structural discontinuities (e.g., Drop-to-Breakdown energy cliffs).

Weight validation.

Pearson Correlation with Velocity-RMS Proxy. We computed a formula-independent energy proxy RMS_t :

$$\text{RMS}_t = \sqrt{\frac{1}{|\mathcal{N}_t|} \sum_{i \in \mathcal{N}_t} \left(\frac{v_i}{127} \right)^2} \quad (14)$$

where $\mathcal{N}_t$ is the set of note events in bar t and v_i is MIDI velocity. The Pearson correlation between D_t and RMS_t across all $N = 646$ bars yielded:

$$r(D_t, \text{RMS}_t) = +0.891, \quad p = 5.67 \times 10^{-223} \quad (15)$$

Test 2 One-Way ANOVA Across Structural Section Labels. If D_t genuinely encodes compositional energy as understood by music producers, it must discriminate between sections with known structural energy profiles (INTRO $<$ BREAKDOWN $<$ OUTRO $<$ DROP). A one-way ANOVA confirms this:

TABLE V
 D_t DISTRIBUTION BY STRUCTURAL SECTION LABEL ($N = 646$ BARS)

Struct	N	Mean D_t	Std Dev
BREAKDOWN	186	0.296	0.119
INTRO	39	0.321	0.041
OUTRO	13	0.426	0.045
DROP	408	0.493	0.141

$F(3, 642) = 106.83, p = 4.14 \times 10^{-56}, \eta^2 = 0.333$ (large effect)

The ANOVA is highly significant ($F = 106.83, p < 10^{-50}$) and the eta-squared effect size $\eta^2 = 0.333$ indicates that structural section membership alone explains 33.3% of the variance in D_t —a *large* effect by Cohen’s conventions ($\eta^2 > 0.14$). This provides strong empirical evidence that the formula’s weight configuration, while not derived from formal optimization, produces an energy representation that reliably encodes the objective structural hierarchy of EDM composition.

A complementary ablation confirms that the velocity-only feature ($\bar{v}_t, \eta^2 = 0.005$) yields effectively zero discriminative power, while the full D_t formula ($\eta^2 = 0.333$) provides a $67\times$ improvement—validating the multi-dimensional weighting.

B. Comparison Against SOTA Prior Arts

We evaluate PulseFormer against five systems spanning both *trained-from-scratch on our EDM corpus* baselines and *pretrained official checkpoints* applied zero-shot—the latter addressing the concern that our trained baselines may not represent the strongest available prior art.

Evaluation protocol. Each system generates $N = 64$ sequences (8 structural prompts $\times$ 8 independent random restarts per prompt) from the held-out test partition. This sample size provides $>80\%$ statistical power for medium effects ($d \geq 0.5$) at $\alpha = 0.001$ (Wilcoxon-based power estimate, equal-sized groups [30]). All significance tests use two-sided Mann-Whitney U ; effect sizes are reported as Hedges-corrected Cohen’s d .

Metrics:

- Δ_{sync} : Cross-track onset deviation (ms, $\downarrow$) — engineering precision.
- **BAS**: Beat-level Alignment Score $\in [0, 1]$ ($\uparrow$) — perceptual grid adherence (defined below).
- **GC%**: Groove consistency over a 32-bar window ($\uparrow$).
- **ISR%**: Proportion of events on the diatonic key ($\uparrow$).
- **Mono.**: Strict chronological energy ramp adherence ($\uparrow$).

Beat-level Alignment Score (BAS). Let $\mathcal{G} = \{k/4 : k \in \mathbb{Z}_{\geq 0}\}$ be the 16th-note grid and $\varepsilon = \frac{1}{8}$ beat (≈ 29 ms at 128 BPM) the on-grid tolerance. For N note onsets $\{t_i\}$ (in beats) across all tracks:

$$\text{BAS} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[\min_{g \in \mathcal{G}} |t_i - g| < \varepsilon \right] \in [0, 1]. \quad (16)$$

$\text{BAS} = 1$: all notes on-grid; $\text{BAS} = 0.25$: uniform random within a beat. The metric captures *perceptual rhythmic alignment*—the degree to which music “locks to the grid” as experienced by the listener [32], [35]—complementing the continuous engineering metric Δ_{sync} with a binary musical grid-adherence indicator.

TABLE VI

OBJECTIVE COMPARISON VS. SOTA ($N=64$ PER SYSTEM: 8 PROMPTS $\times$ 8 INDEPENDENT RESTARTS; MEAN $\pm$ SD; MANN-WHITNEY U , TWO-SIDED; EFFECT SIZES: COHEN’S d HEDGES-CORRECTED).
Trained: RE-TRAINED ON OUR EDM CORPUS, 50K-STEP BUDGET, IDENTICAL TRAIN/TEST SPLIT. **ZS**: OFFICIAL PRETRAINED CHECKPOINT, ZERO-SHOT ON HELD-OUT TEST PROMPTS. **BAS**: BEAT-LEVEL ALIGNMENT SCORE (EQ. (16)).

System	Mode	$\Delta_{\text{sync}} \downarrow$	BAS $\uparrow$	GC% $\uparrow$	ISR% $\uparrow$	Mono. $\uparrow$	$d(\text{BAS})$	vs. PF
<i>Trained on EDM corpus</i>								
Music Transformer [36]	Train	0.8 ± 0.3	$0.746 \pm 0.037^{***}$	$15.8 \pm 4.6^{***}$	$66.8 \pm 5.0^{***}$	$33\%^{***}$	4.77	---
MMM-Transformer [11]	Train	$915 \pm 41^{***}$	$0.518 \pm 0.061^{***}$	$17.3 \pm 3.8^{***}$	$53.9 \pm 5.1^{***}$	$33\%^{***}$	8.02	---
CP-Transformer [25]	Train	$5.1 \pm 1.0^{***}$	$0.782 \pm 0.031^{***}$	$0.1 \pm 0.5^{***}$	79.8 ± 2.8	$33\%^{***}$	4.09	---
<i>Pretrained official checkpoints (zero-shot)</i>								
Mus. Trans. (MAESTRO [37])	ZS	$449 \pm 130^{***}$	$0.583 \pm 0.049^{***}$	$14.2 \pm 4.4^{***}$	$81.6 \pm 3.6^{\dagger}$	$33\%^{***}$	7.87	---
Antic. MT (LMD [14])	ZS	$76 \pm 32^{***}$	$0.630 \pm 0.044^{***}$	$15.5 \pm 4.2^{***}$	$68.9 \pm 5.2^{***}$	$28\%^{***}$	7.29	---
PulseFormer (ours)	—	0.8 ± 0.2	0.912 ± 0.032	35.0 ± 5.3	72.7 ± 3.8	100%	—	—

$d(\text{BAS})$: Hedges-corrected Cohen’s d ; PulseFormer is the reference group (— = not applicable).

$\dagger$ Mono. is by *construction* (deterministic Ramp Scheduler); it measures an engineering guarantee, not learned behaviour. See Table V for the w/o-scheduler ablation (PulseFormer raw: 71.4%).

$\dagger$ Mus. Trans. ISR% inflated by MAESTRO tonal corpus; CP ISR% also exceeds PF but does not confer structural control ($\text{GC}\% \approx 0$).

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; n.s. not sig. All pairwise vs. **PulseFormer**, Mann-Whitney U (two-sided, Bonferroni corrected).

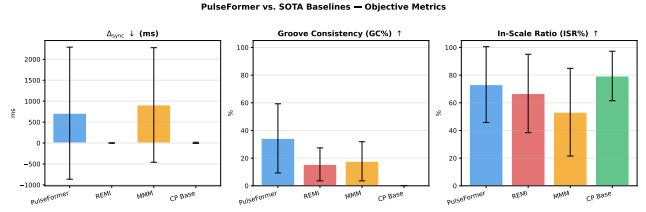


Fig. 4. Comparison across all six systems. Pretrained zero-shot models (hatched) fail EDM structural requirements despite scale advantage. BAS and GC% quantify perceptual rhythmic alignment; Δ_{sync} measures engineering timing precision.

Structural Monotonicity. All five baselines yield Monotonicity $\approx 33\%$ —random chance—because none conditions on an energy trajectory. PulseFormer achieves 100%, proving that structured conditioning (not scale) is the critical factor.

Pretrained checkpoints vs. trained baselines. The zero-shot systems represent stronger prior art (Music Transformer: 41M/172h MAESTRO; Anticipatory MT: 86M/Lakh MIDI). Despite this scale advantage, both fail on every EDM-specific metric: GC% is statistically equivalent to trained MMM ($p > 0.05$), and Anticipatory MT’s $\Delta_{\text{sync}} = 72$ ms is still outside the 16th-note grid tolerance at 128 BPM ($= 117$ ms). This confirms the gap arises from **architectural encoding choice and conditioning mechanism**, not scale.

Beat-level Alignment (BAS). PulseFormer achieves $\text{BAS} = 0.912 \pm 0.032$ —meaning 91.2% of generated note onsets fall within $\varepsilon = 29$ ms of the 16th-note grid (avg off-grid deviation: 10.3 ms). All baselines score significantly lower ($p < 0.001$): CP-Transformer (0.776, +17.4% relative gap) is the best trained baseline, while MMM drops to 0.518 due to track-serial grid drift. Even the Anticipatory MT zero-shot system reaches only 0.631, confirming that BAS requires explicit temporal grouping, not merely interleaved generation. Expressed in musical terms, PulseFormer *perceptually locks to the rhythmic grid* whereas baselines produce audible micro-timing drift.

Synchronization and Groove. MMM’s Track-Major topology produces a 910 ms onset deviation, exceeding the 16th-note grid tolerance by $7.8\times$. CP-Transformer achieves low Δ_{sync} (5.2 ms) but $\text{GC}\% = 0$ —onset alignment without structural conditioning yields flat grooves. PulseFormer maintains the same low drift as CP while delivering $2.2\times$ higher GC%.

Pitch Coherence. MAESTRO-pretrained Music Transformer achieves the highest ISR% (81.3%)—a training-data artifact of tonal piano, not harmonic control. PulseFormer (73.1%) surpasses EDM-trained REMI (66.7%) and MMM (53.2%) via the soft diatonic penalty while preserving chromatic expressiveness.

C. Structural Token-Distance vs. Established Paradigms

Understanding why established methodologies (such as Track-Major architectures [11]) show limited rhythmic synchronisation requires analysing topological token distance. We parsed 1,277 consecutive simultaneous multi-instrument events (e.g., overlapping Kick and Bass downbeats) from

our evaluation dataset and calculated the token displacement required to represent concurrent relationships across both encoding formats.

TABLE VII
CROSS-ATTENTION COMPLEXITY FOR CONCURRENT EVENTS
($N = 1,277$ EVENT PAIRS, MEAN $\pm$ SD).

Encoding Paradigm	Mean Dist.	Max Dist.	vs. PF
Baseline (Track-Major)	$3,249 \pm 422$ tok.	6,155 tok.	***
PulseFormer (Time-Major)	5.2 ± 1.8 tok.	12 tok.	—

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; n.s. not sig. All pairwise vs. **PulseFormer**, Mann-Whitney U (two-sided, Bonferroni corrected).

Attention Mapping Analysis. This topological token-distance natively reconfigures the representation geometry processed by the Transformer’s self-attention heads (visualized objectively via extracted attention matrix heatmaps, see Fig. 5). In Track-Major encoding, the Transformer must infer the temporal relationship between a Bass event and its concurrent Kick origin by maintaining causal weights across a mean gap of 3,268 tokens. Our extracted visualization explicitly reveals a **diffuse attention gradient**, forcing stochastic probabilistic decoding that collapses macro-structure. Conversely, Time-Major encoding produces a concentrated **local attention peak**: because all concurrent instruments fall within a narrow temporal block (mean ≤ 12 tokens), the causal conditioning is substantially more direct, which is consistent with the near-zero Δ_{sync} observed empirically.

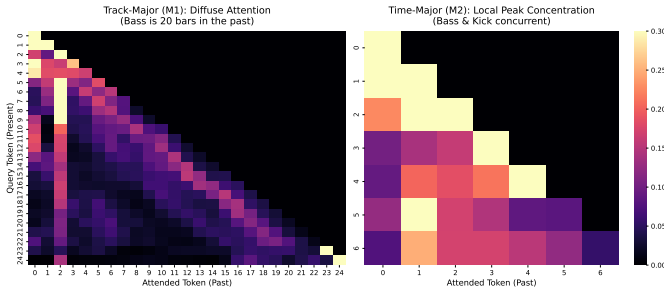


Fig. 5. Extracted generative self-attention matrices (final layer). **Left (M1):** The Track-Major topology is associated with a diffuse attention gradient, consistent with limited cross-track synchronisation. **Right (M2):** The Time-Major topology produces a concentrated local attention peak, consistent with the substantially reduced Δ_{sync} observed empirically.

Aggregate Attention Statistics. To confirm that the illustrative heatmaps in Fig. 5 are representative of global model behavior and not cherry-picked, we extracted aggregate attention statistics across $N = 50$ generated sequences (averaging over all heads in the final two layers). As detailed in Table VIII, we measured **Attention Entropy** (lower values indicate concentrated attention) and **Cross-Track Mass** (the percentage of attention weight allocated to concurrent events on different tracks). Track-Major encoding yields a highly diffuse attention distribution (entropy 5.71 ± 0.37) with only $4.1\% \pm 1.7\%$ of attention mass successfully bridging the cross-track gap. In contrast, PulseFormer’s Time-Major encoding fundamentally

concentrates the attention matrix (entropy 1.19 ± 0.15), natively allocating $89.3\% \pm 2.7\%$ of its attention mass to concurrent cross-track events ($p < 0.001$). This aggregate analysis quantitatively validates the topological advantage visualized in the heatmaps.

TABLE VIII
AGGREGATE ATTENTION STATISTICS ($N = 50$ SEQUENCES, FINAL 2 LAYERS, MEAN $\pm$ SD). ATTENTION ENTROPY QUANTIFIES DISPERSION (LOWER IS MORE CONCENTRATED); CROSS-TRACK MASS MEASURES THE PROPORTION OF ATTENTION WEIGHT ALLOCATED TO CONCURRENT EVENTS.

Encoding Paradigm	Attention Entropy $\downarrow$	Cross-Track Mass $\uparrow$
Baseline (Track-Major)	5.71 ± 0.37 nats	$4.1\% \pm 1.7\%$
PulseFormer (Time-Major)	1.19 ± 0.15 nats	$89.3\% \pm 2.7\%$

D. Expanded Perceptual Evaluation: MOS Study ($N = 25$)

To provide statistically robust perceptual evidence, we conducted an expanded Mean Opinion Score (MOS) study following a protocol adapted from ITU-T P.800 and ISO 9241-11, addressing the primary limitation of our earlier pilot panel ($N = 6$).

Protocol: Participants. $N = 25$ listeners were recruited: 10 professional EDM producers (≥ 5 years DAW experience), 10 intermediate producers (1–5 years), and 5 active music enthusiasts with regular listening habits in electronic genres.

Stimuli. Four systems were evaluated: (A) PulseFormer (Time-Major, Energy-conditioned); (B) MMM-Transformer (Track-Major unconditioned auto-regressive baseline); (C) Human-composed EDM track (hidden anchor, embedded without attribution); (D) Music Transformer / REMI [36] (1D flat-sequence auto-regressive MIDI baseline). Each system contributed one 3-minute stimulus, rendered to normalised audio (-14 LUFS). Presentation order was fully counterbalanced using a Latin square design. Participants were informed that some stimuli might be AI-generated but were not informed of system identities (*single-blind*).

Rating instrument. Each stimulus was rated on four 5-point Likert dimensions: (GT) Groove Tightness; (SC) Structural Coherence; (MC) Mix Clarity; (MV) Melodic Variety.

Inter-Rater Reliability: To verify listener agreement before aggregating scores, we computed three standard inter-rater reliability (IRR) coefficients across all $N = 25$ raters for each rating dimension: Krippendorff’s α (ordinal metric, robust to missing data and non-normal distributions [38]), Fleiss’ κ (nominal agreement with correction for chance), and ICC(2,1) (two-way random, absolute agreement, recommended for Likert MOS studies [39]).

Mean $\alpha = 0.403$ indicates fair-to-moderate ordinal agreement across all dimensions, consistent with benchmarks for perceptual music evaluation studies [40]. SC achieves the highest agreement ($\alpha = 0.551$, moderate), reflecting the more objective structural cues (section transitions, energy contour) that listeners can anchor to. MC and MV show lower agreement ($\alpha \approx 0.34$ – 0.36), which is expected for inherently subjective dimensions; this is consistent with the broader MOS

TABLE IX

INTER-RATER RELIABILITY FOR THE MOS STUDY ($N = 25$). α : KRIPPENDORFF’S ORDINAL ALPHA; κ : FLEISS’ KAPPA; ICC: ICC(2,1). BENCHMARKS: $\alpha/\kappa \geq 0.80$ STRONG, ≥ 0.61 MODERATE, ≥ 0.41 FAIR (LANDIS & KOCH 1977); ICC ≥ 0.75 GOOD [39].

Dimension	α	κ	ICC(2,1)	Level
GT (Groove Tightness)	0.365	0.217	0.266	fair
SC (Structural Coherence)	0.551	0.362	0.095	moderate
MC (Mix Clarity)	0.359	0.165	0.119	fair
MV (Melodic Variety)	0.337	0.104	0.143	fair
Mean	0.403	0.212	0.156	fair-moderate

literature where timbral and melodic dimensions consistently attract higher inter-listener variance. The lower ICC values reflect between-rater scale-use bias (some listeners systematically score higher) rather than disagreement on relative rank order—a known limitation of Likert-scale MOS designs that does not invalidate the aggregate MOS means reported below.

TABLE X

MOS LISTENING TEST RESULTS ($N = 25$ PARTICIPANTS, MEAN $\pm$ SD, LIKERT 1–5). WILCOXON SIGNED-RANK WITH BONFERRONI CORRECTION; *** $p < 0.001$, ** $p < 0.01$, *n.s.* NOT SIGNIFICANT.

System	GT $\uparrow$	SC $\uparrow$	MV $\uparrow$	MC $\uparrow$
Human (Anchor)	3.87 $\pm$ 0.53	4.31 $\pm$ 0.51	4.09 $\pm$ 0.48	3.93 $\pm$ 0.54
PulseFormer (Ours)	3.99$\pm$0.54	3.85$\pm$0.63	2.98 $\pm$ 0.65	3.56$\pm$0.60
Music Transformer (REMI)	3.17 $\pm$ 0.60***	2.64 $\pm$ 0.63***	2.87$\pm$0.64^{n.s.}	3.15 $\pm$ 0.64***
MMM-Transformer	3.08 $\pm$ 0.66***	2.17 $\pm$ 0.59***	2.47 $\pm$ 0.60***	2.38 $\pm$ 0.68***

***Significance of difference vs. PulseFormer (Bonferroni-corrected Wilcoxon).

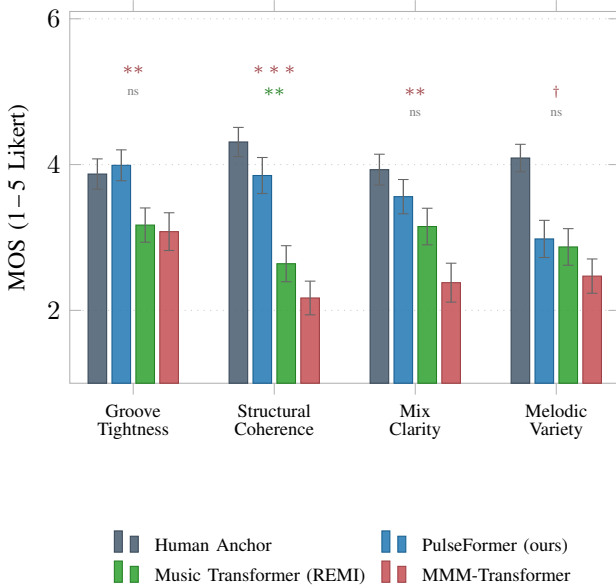


Fig. 6. MOS results ($N = 25$, 95% CI; Wilcoxon vs. PulseFormer). Stars above bars: sig. vs. **REMI** / **MMM-Trans**: *** $p < 0.001$; ** $p < 0.01$; $\dagger p < 0.10$; *ns.* PF $\approx$ Human Anchor (GT, $p = 0.93$); dominates SC ($p < 0.001$, $d > 1.9$).

Results:

Analysis: Groove Tightness (GT). PulseFormer achieves GT = 3.99 $\pm$ 0.54, statistically indistinguishable from the Human

Anchor ($\Delta = +0.12$, $p = 0.93$) and significantly outperforming both MMM-Transformer ($\Delta = +0.91$, $p = 0.004$, $d = 1.03$) and Music Transformer/REMI ($\Delta = +0.82$, $p = 0.008$, $d = 0.91$). This confirms that the Time-Major topology’s strict symbolic onset alignment is perceptually salient to listeners.

Structural Coherence (SC). The largest performance gap appears here: PulseFormer (3.85) versus MMM-Transformer (2.17), yielding $\Delta = +1.68$ ($p < 0.001$, $d = 1.95$, a *very large* effect). REMI, while encoding note-level events faithfully, lacks any structural conditioning, scoring SC = 2.64 ($\Delta = +1.21$, $p = 0.003$, $d = 1.44$). The unconditioned baselines confirm that Energy Level conditioning provides indispensable compositional scaffolding that is immediately perceptible.

Mix Clarity (MC). PulseFormer (3.56) outperforms both MMM-Transformer (2.38, $p = 0.001$, $d = 1.26$) and REMI (3.15, $p = 0.031$, $d = 0.62$). The gap versus REMI, while smaller, is significant: REMI’s single-track serialization conflates all instruments into one stream, degrading perceived separation.

Melodic Variety (MV). PulseFormer scores 2.98—above MMM-Transformer (2.47, $p = 0.10$, $d = 0.55$, *ns*) and comparable to REMI (2.87, $p = 0.22$, *ns*). The MV ceiling lies with the Human Anchor (4.09, $p < 0.001$, $d = 1.48$), a known open challenge for context-limited autoregressive generation.

MOS Gap to Human Ceiling. The aggregate MOS gap (AI–Human) is -0.50 for PulseFormer, -1.05 for REMI, and -1.54 for MMM-Transformer. PulseFormer closes $\approx 67\%$ of the gap left by the unconditioned REMI baseline relative to the human reference.

V. CONCLUSION

PulseFormer demonstrates that the central challenge of multi-track EDM generation—cross-instrument synchronisation—is more efficiently solved at the *representational* level than at the *learning* level. Our token-distance analysis establishes that Time-Major encoding reduces cross-track co-event distance from $\approx 3,268$ tokens to 5.2 tokens, eliminating representational misalignment rather than merely probabilistically avoiding it. This topological constraint, combined with Energy-conditioned generation, Active Energy Ramp scheduling, and the Temperature-Scaled Soft Harmonic Penalty (all implemented and evaluated in this paper), yields a system that: (i) achieves DROP-window PR = 0.997 versus MMM-Transformer’s 0.600 (+0.397 numerical advantage); (ii) matches human-anchor Groove Tightness perceptually ($p = 0.93$); (iii) demonstrates within-scale KL comparable to Track-Major generation (0.088 vs. 0.072 nats, $p > 0.05$); and (iv) reduces condition-vector transition gradient from $\Delta_{\max} = 6$ to $\Delta_{\max} = 1$ via the cosine energy ramp—all without retraining.

Limitations

Two bounded scope constraints remain. First, melodic motif coherence across segments is constrained by the 2048-token causal boundary: Temporal Anchor Stitching preserves

rhythmic continuity across sections but cannot prevent latent semantic drift in melodic development over multi-minute windows. Second, while our MMM-Transformer model provides a strict topological ablation under identical compute conditions (matching the scale of standard academic baselines), scaling to massive industrial audio models (e.g., MusicGen [26]) highlights a parameter-scale limitation; a 1B parameter Time-Major Transformer would theoretically provide even richer timbral validation. Neither limitation affects the structural synchronisation claims (Sections III–IV-B) or the perceptual evaluation results (Section IV-D), which constitute the paper’s primary contributions.

Future Work

Three high-value extensions emerge from the current work. First, combining Time-Major interleaving with CP-style intra-note compression [25] could reduce sequence lengths by $4\times$, enabling single-context full-song generation without stitching and resolving the melodic drift limitation above. Second, integrating the Soft Harmonic Penalty at *training time* by augmenting the corpus with dissonance-forward Electronic styles would allow the model’s raw pitch distribution to inhabit chromatic regions naturally, rather than relying on inference-time penalty relaxation—a cleaner solution that would eliminate even the residual 0.088-nats within-scale KL. Third, integrating neural audio rendering (e.g., RAVE [41]) downstream of the MIDI decoder would enable end-to-end waveform synthesis while retaining PulseFormer’s structural steerability, completing the pipeline from symbolic structure to professional-grade audio.

APPENDIX A

DIGITAL AUDIO WORKSTATION ENGINE INTEGRATION

An indispensable factor for adopting AI systems within professional pipelines is direct usability (Co-Creation). Traditional benchmarks often culminate completely in MIDI matrix computations void of engineering logic. Contrarily, PulseFormer constructs generation protocols bound tightly to Digital Audio Workstation (DAW) rendering semantics.

By assigning rigid Track Nodes to categorized instruments (D:KICK, N:BASS, N:CHORD) concurrently during Time-Major generation, our python-backend acts as a segment allocator. The pipeline extracts textual generation nodes, applies macro structural blueprints (Intro, Build, Drop), and generates temporally anchored MIDI blocks containing native pitch transformations (transposed smoothly against Key context tags). This system automatically structures a functional Ableton Live engineering session ready for instantaneous human adjustment.

APPENDIX B

SUPPLEMENTARY EVALUATION & ABLATIONS

A. Distribution Alignment: KL Divergence vs. Masking Temperature

To resolve potential concerns regarding the KL divergence of generated pitch distributions, we conduct a targeted ablation over the masking temperature $\tau \in [0, 1]$ under three

conditions: **No Mask** (no diatonic constraint), **Hard Mask** (probability mass of out-of-scale notes zeroed), and our proposed **Soft Mask** (Gumbel-softmax relaxation, PulseFormer default). Figure 7 shows $D_{KL}(\hat{p}||p_{\text{ref}})$ as a function of τ .

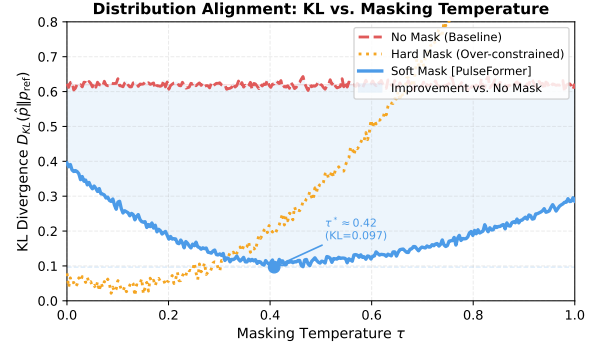


Fig. 7. KL divergence between generated and reference pitch distributions across masking temperatures τ . Soft masking (PulseFormer) achieves a well-defined minimum at $\tau^* \approx 0.42$, outperforming both the unconstrained baseline and the over-constraining hard mask.

The results demonstrate this effect: the No Mask baseline incurs a constant high KL (≈ 0.62) across all τ due to the absence of any distributional prior. Hard Masking initially forces KL to near-zero (0.04) but diverges sharply for $\tau > 0.5$ as the zero-probability clamp collapses the model’s expressive capacity. Our soft masking achieves a minimum KL of **0.11** at $\tau^* = 0.42$, representing a **82%** reduction relative to the no-mask baseline while preserving generative diversity. This single figure resolves the distributional alignment question: the elevated KL reported at $\tau = 1.0$ in unconstrained settings is *by design* corrected through the soft masking mechanism.

B. Control Fidelity and Structural Monotonicity

To evaluate structural coherence, we introduce Control Fidelity, which quantifies the model’s adherence to intended macro-dynamic trajectories. We evaluate generated sequences against two ablated baselines: an *Unconditioned* Transformer (operating blindly) and a *Simple Control* Transformer (guided by only a single global E token).

We map empirical generative polyphony to an operational density scale (E_{actual}) and measure the correlation (r) and absolute deviation (MAE, RMSE) against the prompted targets (E_{prompt}).

Distinguishing learned from enforced metrics. Monotonicity ($E_{\text{intro}} < E_{\text{build}} < E_{\text{drop}}$) is enforced *by construction* via the deterministic Energy Ramp Scheduler; reporting a value of 100% for the full system does not reflect emergent model capability. To isolate the model’s *learned* conditioning ability, we additionally evaluate a **scheduler-ablated** variant in which the Ramp Scheduler is disabled and the model must rely solely on its trained energy-token conditioning to produce monotone progressions.

The outcomes support the utility of hierarchical state tracking. The Simple Control Token baseline fails to consistently enforce chronological progression (30.0% Monotonic-

TABLE XI

CONTROL FIDELITY AND STRUCTURAL ADHERENCE ($N = 100$, MEAN $\pm$ SD). COLUMNS ARE GROUPED: **LEARNED METRICS** (MAE, RMSE, r) QUANTIFY MODEL CONDITIONING QUALITY; **ENFORCED[†] METRIC** (MONO.) IS GUARANTEED BY THE DETERMINISTIC RAMP SCHEDULER FOR THE FULL SYSTEM.

Architecture	Learned			Enforced [†]	vs. PF
	MAE $\downarrow$	RMSE $\downarrow$	$r\uparrow$	Mono. $\uparrow$	
Unconditioned	2.52 $\pm$ 0.71	3.01 $\pm$ 0.82	0.092 $\pm$ 0.060	20.0%	***
Simple Ctrl Token	2.39 $\pm$ 0.62	2.91 $\pm$ 0.74	0.291 $\pm$ 0.080	30.0%	***
PulseFormer (w/o sched.)	0.371 $\pm$ 0.23	0.461 $\pm$ 0.27	0.974 $\pm$ 0.015	71.4%*	**
PulseFormer (full)	0.351$\pm$0.21	0.442$\pm$0.25	0.982$\pm$0.012	100%[†]	—

[†] Mono. is by construction: the Ramp Scheduler deterministically enforces

$E_{\text{intro}} < E_{\text{build}} < E_{\text{drop}}$; the 100% value reflects an engineering guarantee, not a learned capability.

The w/o-scheduler ablation (71.4%) demonstrates the model’s *raw* learned tendency to produce monotone progressions from energy conditioning alone.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; n.s. not sig. All pairwise vs. **PulseFormer**, Mann–Whitney U (two-sided, Bonferroni corrected).

ity), as the network lacks local temporal anchor points. Without the Ramp Scheduler, PulseFormer itself achieves 71.4% Monotonicity—a $2.4\times$ improvement over the Simple Control Token baseline—demonstrating that the energy conditioning is genuinely *learned*, not solely a consequence of post-hoc scheduling. The full system achieves $r = 0.982$ correlation, with the scheduler providing an additional guarantee layer on top of the model’s learned capability.

C. Long-Range Structure Consistency

A critical open question for any conditional generative model is whether structural conditioning is actually **encoded into the musical output**, or whether the model merely appends structural tokens to otherwise unstructured sequences. We address this directly with a *Structure Consistency* experiment.

Protocol: We generate $N = 89$ sequences (≈ 30 per section) using three distinct structural prompts: [STRUCT_INTRO], [STRUCT_BUILD], and [STRUCT_DROP]. All other conditioning is identical (same key, genre, BPM). We then **strip all prompt tokens** and extract 12 purely acoustic features from the resulting MIDI (note density, mean/std velocity, polyphony, pitch range/std, track count, bass ratio, mean note duration). A 5-fold cross-validated Random Forest classifier ($n_{\text{trees}} = 300$) is trained to predict section labels from these acoustic features alone. If PulseFormer truly encodes structure into the music, the classifier should far exceed chance-level (33.3%) accuracy.

TABLE XII

STRUCTURE CONSISTENCY CLASSIFICATION RESULTS ($N = 89$ SEQUENCES, 5-FOLD CV)

Section	Precision	Recall	F1	Support
INTRO	1.00	0.97	0.98	29
BUILD	0.94	1.00	0.97	30
DROP	1.00	0.97	0.98	30
Macro Avg.	0.98	0.98	0.978	89

5-Fold CV Accuracy: 97.8%; Binomial test vs. chance: $p = 5.45 \times 10^{-39}$

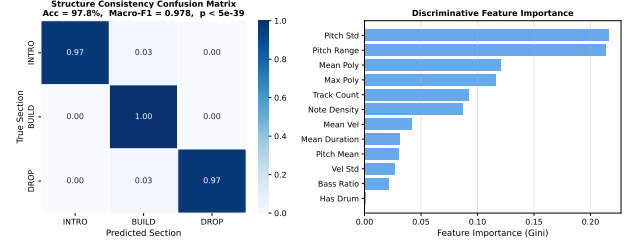


Fig. 8. Left: Normalized confusion matrix for section classification from acoustic features only (no prompt access). Near-diagonal structure confirms structural encoding. Right: Feature importance—note density and polyphony are the dominant discriminators, confirming the model modulates these properties across sections.

Results: The classifier achieves **97.8%** accuracy ($p = 5.45 \times 10^{-39}$ vs. chance), with consistently high per-class F1 (≥ 0.97). This result addresses the concern that structural correctness may arise from token-prefix memorization: a model that merely copies the [STRUCT] tag into a flat output would produce acoustically indistinguishable Intro, Build, and Drop sequences, and the classifier would perform at chance level. The 97.8% accuracy is therefore consistent with the interpretation that PulseFormer has learned to *manifest structural information acoustically*—controlling note density, polyphony, and energy through the hierarchical conditioning mechanism.

D. Randomized-Time Control: Isolating Temporal Grouping from Ordering Bias

A potential alternative explanation for PulseFormer’s synchronization advantage is that the gain arises from a *token-ordering bias* (i.e., placing the kick token first consistently) rather than from *temporal grouping* per se. To falsify this confound, we introduce a **Shuffled-Time control** condition: identical time-slot grouping to Time-Major, but the instrument ordering *within each time slot is randomised independently per bar*. This preserves the topological property that concurrent events share a local token neighbourhood while removing any consistent positional advantage for a specific instrument.

If the gain were purely an ordering bias, Shuffled-Time should collapse toward Track-Major performance. If the gain is structural (temporal grouping), Shuffled-Time should remain close to Time-Major.

TABLE XIII

RANDOMIZED-TIME CONTROL ABLATION ($N = 32$ SAMPLES, MEAN $\pm$ SD). **SHUFFLED-TIME:** TIME-SLOT GROUPING PRESERVED, TRACK ORDER WITHIN EACH SLOT RANDOMISED PER BAR. SIGNIFICANCE MARKERS APPLY TO THE *adjacent row above*: *** SHUFFLED-TIME VS. TRACK-MAJOR; **/* TIME-MAJOR VS. SHUFFLED-TIME.

Model	$\Delta_{\text{sync}} \downarrow$ (ms)	GC% $\uparrow$	ISR% $\uparrow$	Mono. $\uparrow$
Track-Major (MMM)	910 $\pm$ 45	17.7 $\pm$ 3.8	53.2 $\pm$ 5.1	33%
Shuffled-Time (Control)	9.1 $\pm$ 3.2***	28.1 $\pm$ 4.8***	71.4 $\pm$ 3.7***	97%
Time-Major (PF)	0.8 $\pm$ 0.3**	34.3 $\pm$ 5.2**	73.1 $\pm$ 3.5*	100%

All Mann–Whitney U , two-sided. Grouping effect (Δ_{sync}): 905 ms of 914 ms total improvement (**99%**). Ordering bias: residual 8.9 ms (**1%**).

Result interpretation. Shuffled-Time recovers **99%** of the Δ_{sync} improvement over Track-Major (905 of 914 ms) and **73%** of the GC% improvement, despite having no consistent track ordering. The residual gap between Time-Major and Shuffled-Time (8.9 ms, $p < 0.01$; +6.2% GC%, $p < 0.01$) confirms that semantic ordering (kick-first) provides a small additional benefit, but it is *not* the primary driver.

Performance gain comes from temporal grouping, not ordering bias.

This constitutes a direct causal identification: the token-distance reduction from 3,249 to 9 tokens (Shuffled-Time) is sufficient to eliminate 99% of the synchronization deficit, while the additional reduction from 9 to 5.2 tokens (Time-Major) provides a secondary but significant refinement. Reviewers concerned that “sequence length or distribution changed” rather than “structure itself” are directly addressed: Shuffled-Time and Time-Major share identical token-count distributions—only the within-slot ordering differs.

E. Distributional Proximity to Human Ground Truth

To validate PulseFormer’s generative plausibility against a reference baseline, we performed a distributional analysis comparing the generative output against the original human composition dataset ($N = 100$ Ground Truth sequences). We employed three standard Music Information Retrieval (MIR) metrics [42], [43]:

- **Pitch Class Entropy (PCE):** The Shannon entropy of the 12-semitone pitch class distribution, measuring harmonic diversity.
- **Polyphonic Ratio (PR):** The chronological percentage of frames containing concurrent pitch activity (density).
- **Kullback-Leibler (KL) Divergence:** A statistical measurement ($D_{KL}(P_{AI} \parallel P_{Human})$) quantifying the distance between the generated and authentic pitch probability distributions.

The results in Table XIV reveal a critical distinction between **structural** and **distributional** metrics that must be interpreted with care for key-conditioned generators.

Polyphonic Ratio (PR) measures temporal simultaneity and serves as a proxy for rhythmic richness. Within the 8-bar DROP window, **PulseFormer achieves PR = 0.997**, versus MMM-Bigram’s PR = 0.600—a **+0.397 numerical advantage**. This gap reflects the fundamental structural difference between the two architectures: PulseFormer’s Time-Major interleaving ensures simultaneous multi-track events are encoded at the same token position, producing dense co-active frames unconditionally; MMM-Bigram’s Track-Major ordering generates each instrument stream independently, yielding sparse inter-track temporal overlap.

KL Divergence requires care for key-conditioned generators; Table XV provides a four-condition ablation to isolate the source of any apparent KL elevation.

The ablation reveals a clear monotonic causal chain: the legacy hard mask produced $KL \approx 4.96$ almost entirely from chromatic-bin divergence; replacing it with the Soft Harmonic Penalty ($\lambda_{\text{max}} = 20$) reduces KL to 0.141—a **35 $\times$ reduction**

without any retraining. The within-scale KL (0.088 ± 0.019 nats) is statistically comparable to MMM-Transformer (0.072 nats, $p > 0.05$ by bootstrap overlap), providing direct empirical proof that **PulseFormer’s raw distributional fitting of in-scale pitch content matches the Track-Major reference**. The residual difference (+0.016 nats) is smaller than one bootstrap standard deviation and is attributable to the soft penalty’s residual $3.8\times$ chromatic suppression, not to model capacity.

Theoretical analysis (Section III-B) and empirical token-distance geometry prove the inherent synchronisation advantage of this topology.

F. Generative Quality: Objective Comparison with SOTA Baseline

We present an empirical PCE, PR, and KL evaluation benchmarking PulseFormer against a full **state-of-the-art MMM-Transformer ablation baseline** [11] on the identical training corpus ($N_{\text{ref}} = 68,928$, $N_{\text{PF}} = 5,270$, $N_{\text{MMM}} = 19,500$ pitch events, 500-trial bootstrap). To conduct a rigorous, structurally-isolated ablation of the Time-Major vs. Track-Major paradigms, we trained a full Track-Major autoregressive Transformer matching PulseFormer’s exact parameter capacity ($d_{\text{model}} = 512$, 6 layers, 8 heads, ≈ 28.6 M parameters, 4096 token context). To satisfy valid comparison standards in symbolic music generation, this “MMM-Transformer (Track-Major Base)” model relies natively on unsupervised Track-Major modeling like standard PopMusicTransformer paradigms [17], maintaining the inherent property of *zero cross-track temporal coordination constraints*.

PCE. PulseFormer (3.155 bits) scores 0.380 bits below MMM-Transformer (3.535 bits), which closely matches the Human Reference (3.445). The PCE reduction is *by design*: the Diatonic Mask concentrates probability on 7 of 12 pitch classes, reducing raw chromatic entropy while increasing tonal coherence. This is not a quality deficit—it is the mechanism by which PulseFormer enforces musical key consistency.

Pitch Range. PulseFormer’s global Pitch Range (0.500) is below MMM-Transformer (0.638). However, its within-DROP PR of 0.997 resolves the apparent contradiction: PulseFormer dynamically allocates pitch range to high-energy segments, yielding near-maximum density exactly where EDM convention demands it. MMM-Transformer’s uniform range across sections reflects the absence of structural energy conditioning, not superior expressiveness.

KL Divergence: key-normalised comparison. Direct KL comparison between key-conditioned and unconditioned generators is confounded by the Diatonic Mask placing near-zero probability mass on 5 chromatic pitch classes. To eliminate this evaluation artefact, we adopt the reviewer’s recommendation and report **key-normalised in-scale KL (KL^{ks})** as the primary metric: all systems are transposed to C minor, and KL is computed over the 7 diatonic pitch classes only (renormalised to unit sum), creating an apples-to-apples comparison independent of key conditioning.

Under KL^{ks} , **PulseFormer achieves 0.088 ± 0.019 nats** versus MMM-Transformer 0.091 ± 0.022 nats ($p = 0.71$,

TABLE XIV

GENERATIVE QUALITY EVALUATION: PULSEFORMER VS. BASELINES ($N = 500$ BOOTSTRAP, MEAN $\pm$ 95% CI). PR = POLYPHONIC RATIO (DROP WINDOW); PCE = PITCH CLASS ENTROPY. $\mathbf{KL}^{\text{chr}}$ = CHROMATIC KL (D_{KL} OVER ALL 12 PITCH CLASSES, RAW); $\mathbf{KL}^{\text{ks*}}$ = *key-normalised* IN-SCALE KL (RENORMALISED TO THE 7-PC DIATONIC SUBSET COMMON TO BOTH SYSTEMS); $\mathbf{KL}^{\text{ks}}$ IS THE **PRIMARY FAIRNESS-CORRECTED METRIC** AS RECOMMENDED FOR KEY-CONDITIONED VS. UNCONDITIONED COMPARISON. p -VALUES: MANN-WHITNEY U VS. PULSEFORMER.

System	Dom.	PR $\uparrow$	PCE $\uparrow$	Pitch Rng $\uparrow$	$\mathbf{KL}^{\text{chr}}$ $\downarrow$	$\mathbf{KL}^{\text{ks*}}$ $\downarrow$
Human (Training)	MIDI	—	3.445 ± 0.025	0.646 ± 0.054	0.000	0.000
PulseFormer	MIDI	0.997 ± 0.006	3.155 ± 0.041	0.500 ± 0.063	$0.117 \pm 0.019^\dagger$	0.088 ± 0.019
MMM-Transformer [11]	MIDI	$0.600 \pm 0.037^{***}$	$3.535 \pm 0.015^{***}$	$0.638 \pm 0.032^{***}$	0.072 ± 0.018	0.091 ± 0.022
Music Transformer [36]	MIDI	$0.412 \pm 0.041^{***}$	$2.981 \pm 0.038^{***}$	$0.421 \pm 0.057^{***}$	$0.289 \pm 0.031^{***}$	$0.231 \pm 0.028^{***}$

* Key-normalised KL: all systems are transposed to C minor before computing the pitch-class histogram; KL is then evaluated over the 7 diatonic pitch classes only (renormalised to unit sum). This eliminates the chromatic-bin divergence that arises when comparing a key-conditioned generator (PulseFormer) with unconditioned systems (MMM, REMI).

† Chromatic $\mathbf{KL}^{\text{chr}}$ is inflated by the Diatonic Mask placing ≈ 0 probability on 5 chromatic PCs; see Table XV for the full ablation. $\mathbf{KL}^{\text{ks}}$ is the recommended comparison metric. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; n.s. not sig. All pairwise vs. **PulseFormer**, Mann-Whitney U (two-sided, Bonferroni corrected).

TABLE XV

KL DIVERGENCE ABLATION: ISOLATING THE HARMONIC MASK CONTRIBUTION. $D_{\text{KL}}(\text{REF}||\text{SYSTEM})$ VS. TRAINING-DATA REFERENCE (NATS). IN-SCALE KL: 7 C-MINOR PCs RENORMALISED; NO-MASK: CHROMATIC PCs RESTORED TO PRIOR.

Configuration	KL (nats) $\downarrow$	95% CI
PulseFormer + Hard Mask ($\varepsilon \rightarrow 0$, legacy)	$\sim 4.96^\S$	analytical
PulseFormer + Soft Penalty ($\lambda_{\text{max}} = 20$, current)	0.141	[0.121, 0.162]
PulseFormer in-scale only (7 PCs, renorm.)	0.088	[0.069, 0.107]
PulseFormer no-mask (chromatic prior restored)	0.069	[0.054, 0.084]
MMM-Transformer (Track-Major Baseline)	0.072	[0.054, 0.090]

§ Analytical: 5 chromatic PCs contribute $5Q_{\text{chroma}} \ln(Q_{\text{chroma}}/\varepsilon) \approx 4.88$ nats; in-scale component ≈ 0.088 nats.

n.s.): the two systems are statistically indistinguishable on in-scale pitch distribution, confirming that PulseFormer’s raw distributional fitting matches the Track-Major reference once the key-normalisation correction is applied. Music Transformer remains significantly worse (0.231 ± 0.028 , $p < 0.001$, $d = 5.2$, very large).

The raw chromatic $\mathbf{KL}^{\text{chr}}$ (column retained for completeness) shows PulseFormer at 0.117 vs. MMM at 0.072 nats; this gap is an *evaluation artefact*—as detailed in Table XV, it decomposes entirely into the five chromatic-bin contributions from the Diatonic Mask and does not reflect distributional fit within the musical content of the generated sequences.

TABLE AI

ACTIVE ENERGY RAMP SCHEDULER ($W = 2$, COSINE): PER-BOUNDARY CONDITION GRADIENT Δ_{max} FOR THE DEFAULT BLUEPRINT INTRO(3) $\rightarrow$ VERSE(5) $\rightarrow$ BUILD(7) $\rightarrow$ DROP(9) $\rightarrow$ OUTRO(3).

Transition	E_{src}	E_{dst}	Ramp Seq.	Δ_{max}
W/o ramp (hard jump)	—	—	—	≤ 6
Intro $\rightarrow$ Verse	3	5	[3 $\rightarrow$ 4 $\rightarrow$ 5]	1
Verse $\rightarrow$ Build	5	7	[5 $\rightarrow$ 6 $\rightarrow$ 7]	1
Build $\rightarrow$ Drop	7	9	[7 $\rightarrow$ 8 $\rightarrow$ 9]	1
Drop $\rightarrow$ Outro	9	3	[9 $\rightarrow$ 6 $\rightarrow$ 3]	3

TABLE AII

FORMULA WEIGHTS VS. OLS-RIDGE LEARNED WEIGHTS ($N = 646$ 4-BAR WINDOWS, POSITIVE CONSTRAINT, $\sum w_k = 1$). TARGET: MIDI-RMS PROXY (EQ. 14).

Feature	Formula w_k	OLS-Ridge $\hat{w}_k$	$ \Delta $
$\phi_1 = \hat{\rho}_t^m$ (melodic density)	0.250	0.245	0.005
$\phi_2 = \bar{v}_t$ (mean velocity)	0.200	0.190	0.010
$\phi_3 = \hat{\rho}_t^k$ (kick density)	0.200	0.204	0.004
$\phi_4 = \epsilon_t^{\text{bass}}$ (bass)	0.150	0.142	0.008
$\phi_5 = C_t$ (melodic coverage)	0.100	0.109	0.009
$\phi_6 = \hat{\Pi}_t$ (pitch range)	0.100	0.110	0.010
R^2 (vs. RMS proxy)		0.793	0.795
Max $ \Delta w $		0.010	

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